BRAC University

Department of Computer Science and Engineering

CSE422: Artificial Intelligence

Lab Project Report

Diabetes Prediction Using Machine Learning

Dataset	Diabetes Dataset (100,000 records)
Problem Type	Binary Classification
Models Used	Decision Tree, Logistic Regression, Neural Network, KMeans
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1. Introduction

Diabetes is one of the most prevalent chronic diseases globally, affecting hundreds of millions of people and placing an enormous burden on healthcare systems. Early detection is critical — it allows for timely medical intervention, lifestyle adjustments, and better long-term patient outcomes. Traditional diagnosis requires laboratory tests and clinical consultations, which may not always be accessible or timely.

This project aims to build and evaluate machine learning models that can predict whether a patient has diabetes based on a set of clinical and demographic features. The motivation is to demonstrate how data-driven approaches can supplement clinical decision-making, particularly in resource-limited settings where medical infrastructure is insufficient.

The project follows the complete machine learning workflow: data loading, exploratory data analysis, preprocessing, model training, and evaluation. Three supervised models — Decision Tree, Logistic Regression, and a Neural Network — are trained and compared. Additionally, KMeans clustering is applied to treat the problem as an unsupervised task and explore natural groupings within the data.

2. Dataset Description

2.1 Dataset Overview

The dataset used in this project is the Diabetes Prediction Dataset, which contains clinical and lifestyle information for 100,000 patients.

Property	Value
Number of Records	100,000
Number of Features	8 input features + 1 target
Target Variable	diabetes (0 = No Diabetes, 1 = Diabetes)
Problem Type	Binary Classification
Source	diabetes_dataset.csv

Why Classification? The target variable 'diabetes' is a discrete binary label (0 or 1), making this a classification problem. There is no continuous output to predict, ruling out regression.

Feature Types:

Feature	Type	Category	Description
age	Float	Quantitative	Patient age in years (0.08 – 80)
bmi	Float	Quantitative	Body Mass Index
HbA1c_level	Float	Quantitative	Glycated hemoglobin level

blood_glucose_level	Float	Quantitative	Blood glucose reading
hypertension	Integer (0/1)	Quantitative (Binary)	Has hypertension or not
heart_disease	Integer (0/1)	Quantitative (Binary)	Has heart disease or not
gender	String	Categorical	Female / Male / Other
smoking_history	String	Categorical	never/current/former/ ever/not current/No Info
diabetes	Integer (0/1)	Target	Diabetic (1) or Not (0)

Encoding Requirement: Yes. The features 'gender' and 'smoking_history' are categorical text variables that machine learning algorithms cannot process directly. They must be encoded into numerical form. Label Encoding is applied, converting each unique string value to an integer.

2.2 Correlation Analysis

A Pearson correlation analysis was performed on all features (after temporarily encoding categoricals). The results are shown below:

Feature	Correlation with Diabetes
blood_glucose_level	0.418
HbA1c_level	0.401
age	0.258
bmi	0.214
hypertension	0.198
heart_disease	0.171
smoking_history	0.078
gender	0.020
diabetes (self)	1.000

Observations from Correlation Analysis: blood_glucose_level and HbA1c_level have the strongest positive correlations with the target variable (0.418 and 0.401 respectively). This is clinically meaningful — both are primary diagnostic indicators for diabetes. age and bmi show moderate positive correlations (~0.21–0.26), consistent with known diabetes risk factors. hypertension and heart_disease show weak-to-moderate correlations (~0.17–0.20), suggesting comorbidities have some predictive value. gender has a near-zero correlation (0.020), meaning it has minimal predictive power for diabetes in this dataset. smoking_history has a very weak correlation (0.078).

2.3 Imbalanced Dataset

The class distribution of the target variable 'diabetes' is highly imbalanced:

Class	Count	Percentage
0 (No Diabetes)	82,284	91.5%
1 (Diabetes)	7,634	8.5%

Total	89,918	100%

The dataset is highly imbalanced with a ratio of approximately 10.8:1 (No Diabetes : Diabetes). This means a naive model that always predicts 'No Diabetes' would achieve ~91.5% accuracy — which is misleading. Due to this imbalance, Stratified Splitting is used during train/test split to ensure both classes are proportionally represented. Recall (sensitivity) for the diabetic class is a more important metric than accuracy alone.

A bar chart of class distribution is included in the notebook (Section 3.2).

2.4 Exploratory Data Analysis

EDA was performed to extract important relationships from the data. Key findings:

- HbA1c_level and blood_glucose_level show clear bimodal distributions when split by diabetes class — diabetic patients cluster at higher values.
- Older patients (age > 50) have significantly higher rates of diabetes compared to younger patients.
- Higher BMI values correlate with increased diabetes prevalence, particularly BMI > 30 (obese range).
- Patients with hypertension show a notably higher proportion of diabetes (~30%) compared to those without hypertension (~8%).
- Similarly, patients with heart disease show elevated diabetes rates (~20% vs ~8%).
- Gender shows minimal difference in diabetes rates across Male, Female, and Other categories.
- Current and former smokers show marginally higher diabetes rates than never-smokers.

The pairplot of key numeric features reveals clear clustering patterns, especially in the HbA1c_level vs blood_glucose_level plane, where diabetic patients form a distinct upper-right cluster.

3. Dataset Pre-processing

3.1 Fault: Missing / Null Values

Problem: The dataset contains missing values across all 9 columns. The missing value counts before preprocessing are:

Feature	Missing Values
gender	9,835
age	10,065
hypertension	10,148
heart_disease	9,919
smoking_history	9,988
bmi	9,944
HbA1c_level	10,139
blood_glucose_level	9,880

diabetes	10,082

Solution: All rows containing any null/missing value were dropped (listwise deletion). This approach was chosen because: (1) missing values appear simultaneously across multiple columns in the same rows, making imputation unreliable; (2) with 100,000 original rows, dropping incomplete records still leaves 38,636 clean rows — sufficient for robust model training. After dropping nulls, no missing values remain in any column.

Stage	Row Count	Missing Values
Before	100,000	~10,000 per column
After	38,636	0

3.2 Fault: Categorical Variables

Problem: Two features — gender (Female/Male/Other) and smoking_history (never/current/former/ever/not current/No Info) — are text-based categorical variables. Machine learning algorithms require numerical inputs and cannot process raw strings.

Solution: Label Encoding was applied to both categorical columns. Each unique string value is mapped to an integer. This is appropriate here because: (1) tree-based models (Decision Tree) handle label-encoded categoricals well without assuming ordinality; (2) the number of unique categories is small (3 for gender, 6 for smoking_history), so the integer representation does not introduce significant bias.

Feature	Label Encoding Mapping
gender	Female=0, Male=1, Other=2
smoking_history	No Info=0, current=1, ever=2, former=3, never=4, not current=5

3.3 Fault: Feature Scaling

Problem: Features have vastly different numerical ranges. For example, blood_glucose_level ranges from 80 to 300, while hypertension is binary (0 or 1). Distance-based and gradient-based models (Logistic Regression, Neural Network) are highly sensitive to feature scale.

Solution: StandardScaler (Z-score normalization) was applied — transforming each feature to have zero mean and unit variance. Critically, the scaler was fit only on training data and then applied to test data, preventing data leakage.

4. Dataset Splitting

Stratified splitting was used due to the significant class imbalance (10.8:1 ratio). Stratification ensures that both the training and test sets contain the same proportion of diabetic and non-diabetic patients as the full cleaned dataset.

Split	Samples	Percentage	Class Distribution

Training Set	30,908	80%	~2,615 diabetic / ~28,293 non-diabetic
Test Set	7,728	20%	654 diabetic / 7,074 non-diabetic
Total	38,636	100%	~3,269 diabetic / ~35,367 non-diabetic

StandardScaler was fit on training data only and then applied to both training and test sets to prevent data leakage.

5. Model Training & Testing

5.1 Decision Tree Classifier

Configuration: max_depth=8, random_state=42. The maximum depth is limited to 8 to prevent overfitting.

Class	Precision	Recall	F1-Score	Support
No Diabetes	0.97	1.00	0.98	7,074
Diabetes	0.96	0.67	0.79	654
Accuracy			0.9702	7,728
Macro Avg	0.97	0.84	0.89	7,728
Weighted Avg	0.97	0.97	0.97	7,728

5.2 Logistic Regression

Configuration: max_iter=1000, random_state=42. A linear baseline classifier well-suited to binary classification.

Class	Precision	Recall	F1-Score	Support
No Diabetes	0.96	0.99	0.98	7,074
Diabetes	0.87	0.61	0.72	654
Accuracy			0.9590	7,728
Macro Avg	0.92	0.80	0.85	7,728
Weighted Avg	0.96	0.96	0.96	7,728

5.3 Neural Network (MLPClassifier)

Architecture: Input(8) → Hidden Layer 1 (128 neurons, ReLU) → Hidden Layer 2 (64 neurons, ReLU) → Hidden Layer 3 (32 neurons, ReLU) → Output (Sigmoid). Early stopping was enabled with a validation fraction of 10%. Training stopped after 42 iterations.

Class	Precision	Recall	F1-Score	Support
No Diabetes	0.97	1.00	0.98	7,074

Diabetes	0.96	0.67	0.79	654
Accuracy			0.9696	7,728
Macro Avg	0.97	0.83	0.89	7,728
Weighted Avg	0.97	0.97	0.97	7,728

5.4 KMeans Clustering (Unsupervised)

The dataset was treated as an unsupervised learning problem. KMeans clustering was applied to discover natural groupings in the data without using the target labels.

Elbow Method: The elbow method was used to determine the optimal number of clusters. Inertia was computed for k=2 through k=10. The elbow curve shows a significant bend at k=2, confirming that 2 clusters is the most natural grouping — consistent with the binary nature of the target.

k Value	Observation
k=2	Significant bend — optimal
k=3	Marginal improvement
k=4–10	Diminishing returns

KMeans with k=2 was fitted on the training data. PCA (Principal Component Analysis) was used to reduce the 8-dimensional feature space to 2 dimensions for visualization. The PCA explained variance ratios were: PC1=22.9%, PC2=14.5% (total 37.4%).

Cluster vs True Labels: Comparing the KMeans cluster assignments with the true diabetes labels shows that the two clusters roughly correspond to the diabetic and non-diabetic groups. This validates the inherent separability of the data and confirms that clinical features alone (without labels) can naturally separate diabetic from non-diabetic patients.

6. Model Comparison & Evaluation

Overall Performance Summary:

Model	Accuracy	Precision (Diabetes)	Recall (Diabetes)	AUC
Decision Tree	97.02%	96.49%	67.28%	0.9712
Logistic Regression	95.90%	86.71%	60.86%	0.9561
Neural Network	96.96%	96.45%	66.51%	0.9721

6.1 Accuracy Comparison

All three models achieved accuracy above 95%. Decision Tree (97.02%) and Neural Network (96.96%) performed nearly identically in accuracy, while Logistic Regression (95.90%) was slightly lower.

Bar chart of model accuracy is in the notebook (Section 7.1).

6.2 Precision & Recall Comparison

Precision (for the Diabetes class) measures how often a 'Diabetes' prediction is correct. Recall (Sensitivity) measures how many actual diabetes cases are correctly identified.

Model	Precision	Recall	Interpretation
Decision Tree	96.49%	67.28%	The model rarely gives a false positive for diabetes, but misses ~33% of actual cases.
Logistic Regression	86.71%	60.86%	Lower precision and recall — the linear boundary struggles with non-linear patterns.
Neural Network	96.45%	66.51%	Very similar to Decision Tree. High precision, moderate recall.

Precision/Recall bar chart is in the notebook (Section 7.2).

6.3 Confusion Matrices

Confusion matrices for all three models on the test set (7,728 samples: 7,074 No Diabetes, 654 Diabetes):

Model	True Negatives	False Positives	False Negatives	True Positives
Decision Tree	7,058	16	214	440
Logistic Regression	7,013	61	256	398
Neural Network	7,058	16	219	435

Key Observation: All models have very few False Positives (16–61), meaning they rarely misclassify a healthy patient as diabetic. However, False Negatives are significant (214–256), meaning many actual diabetic patients are missed. In a clinical context, minimizing False Negatives (maximizing Recall) is critically important.

6.4 ROC Curves & AUC Scores

The ROC (Receiver Operating Characteristic) curve plots True Positive Rate vs False Positive Rate at various classification thresholds. AUC (Area Under the Curve) summarizes the model's discriminative ability — closer to 1.0 is better.

Model	AUC Score	Interpretation
Decision Tree	0.9712	Excellent discriminative ability
Logistic Regression	0.9561	Very good — strong baseline
Neural Network	0.9721	Best overall — highest AUC
Random Classifier	0.5000	Baseline reference

ROC curve plot is in the notebook (Section 7.4).

The Neural Network achieves the highest AUC (0.9721), indicating the best overall ability to distinguish diabetic from non-diabetic patients across all thresholds. Decision Tree is a very close second (0.9712). Logistic Regression, while slightly behind, still demonstrates strong performance (0.9561) as a linear model.

7. Conclusion

7.1 Understanding the Results

All three supervised models achieved high accuracy (>95%) and excellent AUC scores (>0.95). The primary discriminating features — HbA1c_level and blood_glucose_level — are the strongest predictors, as confirmed by both the correlation analysis and the model performance. The Neural Network achieved the highest AUC (0.9721), demonstrating its ability to learn complex non-linear boundaries across 8 features simultaneously.

7.2 Model Performance Comments

- Neural Network (Best AUC: 0.9721): Learns non-linear decision boundaries effectively. With only 42 training iterations (early stopping), it converged quickly without overfitting. Highest overall discriminative performance.
- Decision Tree (Best Accuracy: 97.02%): Interpretable and competitive. High precision (96.49%) means it rarely makes false 'Diabetes' predictions. The depth limit (max_depth=8) controls overfitting.
- Logistic Regression (Fastest Training): A solid linear baseline. Lower precision (86.71%) and recall (60.86%) compared to tree/NN models, suggesting the diabetes decision boundary has non-linear components that a linear model cannot fully capture.
- KMeans Clustering: Successfully identified two natural data clusters that roughly align with the diabetic/non-diabetic split — validating that clinical features alone contain sufficient information to separate the two groups without using labels.

7.3 Why These Results?

- High accuracy is primarily driven by two dominant features: HbA1c_level and blood_glucose_level. These clinical markers are direct diagnostic indicators, giving models a strong signal to learn from.
- Recall is moderate (~60–67%) across all models due to class imbalance — with 10.8x more negative samples, models are biased toward predicting 'No Diabetes'. Only 8.5% of the data is diabetic, so the model learns a conservative boundary.

- The Neural Network and Decision Tree outperform Logistic Regression because the true decision boundary is non-linear — patients with moderate blood glucose AND high HbA1c are more likely diabetic, which is a non-linear interaction.

7.4 Challenges Faced

- High Missing Value Rate (~10% per column): Approximately 61,364 rows were removed during preprocessing. While sufficient data remained, this significant reduction may have introduced sampling bias. Imputation was considered but rejected due to simultaneous missingness across columns.
- Severe Class Imbalance (10.8:1): The dominant class (No Diabetes) suppresses recall for the minority class (Diabetes). In a real clinical deployment, techniques like SMOTE oversampling or class-weighted loss functions would be critical to improve recall.
- KMeans Label Alignment: KMeans cluster labels are arbitrary integers. PCA visualization was required to interpret whether clusters correspond to the actual diabetic/non-diabetic groups.
- Neural Network Training Time: Training MLPClassifier on ~30,000 samples with 3 hidden layers is computationally intensive. Early stopping was essential to reduce training time without sacrificing performance.